

Assessment Book — Electrostatics (charges, Coulomb's law, electric field, flux, Gauss's law)

Physics | Grade 11-12

Questions

Total marks 74

Q1. (4 marks) Two point charges are located in the xy-plane: charge $q_1 = +3.0 \mu\text{C}$ is at $r_1 = (0.00, 0.00, 0.00) \text{ m}$ and charge $q_2 = -2.0 \mu\text{C}$ is at $r_2 = (0.30, 0.40, 0.00) \text{ m}$. Using Coulomb's law in vector form, calculate the electrostatic force (vector) on q_2 due to q_1 . Give your answer in newtons with components F_x and F_y to three significant figures.

Working:

Q2. (1 mark) A test charge $q_0 = +1.0 \mu\text{C}$ sits at the origin ($x = 0$). Two other point charges lie on the x-axis: $q_1 = +2.0 \mu\text{C}$ at $x = +0.20 \text{ m}$ and $q_2 = -3.0 \mu\text{C}$ at $x = -0.30 \text{ m}$. Using Coulomb's constant $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$ and the principle of superposition, what is the net electrostatic force on q_0 (give magnitude and direction along the x-axis)?

- A. 0.75 N directed to the left (negative x-direction)
- B. 0.75 N directed to the right (positive x-direction)
- C. 0.30 N directed to the left (negative x-direction)
- D. 0.45 N directed to the left (negative x-direction)

Q3. (3 marks) Use Gauss's law to derive the magnitude of the electric field in each case. Show the Gaussian surface you choose, write the flux = enclosed charge/ ϵ_0 relation, and solve for E . (a) At distance r from an infinitely long straight wire with linear charge density λ . (b) Near an infinite uniformly charged plane with surface charge density σ . (c) Outside ($r > R$) and inside ($r < R$) a uniformly charged thin spherical shell of total charge q and radius R .

Q4. (6 marks) When you rub a plastic comb on dry hair the comb becomes negatively charged and the hair becomes positively charged. Using an atomic-scale description of charge transfer, explain why rubbing transfers electrons between the two objects but does not create or destroy net electric charge. In your answer include how the conservation and quantisation of charge apply to this process. (6 marks)

Q5. (4 marks) A simple electric dipole consists of two point charges $+q$ and $-q$ placed on the z-axis at $z = +a$ and $z = -a$ respectively. Let p denote the dipole moment vector $p = 2a q \hat{z}$ (pointing from $-q$ to $+q$). Assuming the field point is at distance r from the origin with $r \gg a$, (a) derive the far-field electric field on the positive z-axis ($\theta = 0$) and express it in terms of p ; (b)

derive the far-field electric field at a point on the equatorial plane ($\theta = 90^\circ$) at distance r from the origin and express it in terms of p . Give your answers in SI units and keep factors of $1/(4\pi\epsilon_0)$ explicit. (Total 4 marks)

Working:

Q6. (1 mark) Which of the following gives the correct definitions and SI units for the linear, surface, and volume charge densities (λ , σ , ρ)?

- A. λ is charge per unit length (C/m), σ is charge per unit volume (C/m³), ρ is charge per unit area (C/m²).
- B. λ is charge per unit length (C/m³), σ is charge per unit area (C/m²), ρ is charge per unit volume (C/m).
- C. λ is charge per unit length (C/m), σ is charge per unit area (C/m²), ρ is charge per unit volume (C/m³).
- D. λ is total charge divided by cross-sectional area (C/m²), σ is charge per unit length (C/m), ρ is charge per unit volume (C/m³).

Q7. (3 marks) Give the operational definition of the electric field at a point in space. In your answer, state how you would measure it with a test charge, provide the formula that relates the measured quantity to the electric field (use symbols), and give the SI unit of the electric field. (3 marks)

Q8. (6 marks) Explain why the electric field inside a hollow conducting sphere is zero when the conductor is in electrostatic equilibrium, and why any excess charge placed on the conductor resides on its outer surface. In your explanation, apply Gauss's law and describe what happens to the distribution of charges and the electric field if a point charge is placed inside the hollow cavity (not touching the conductor).

Q9. (4 marks) A small metal sphere A is given a total charge of $+8.0 \mu\text{C}$. It is then touched to an identical, uncharged metal sphere B and the two spheres are separated.

(a) Calculate the final charge on each metal sphere after they are separated. (2 marks)

(b) If sphere B had instead been an identical neutral plastic (insulating) sphere and you repeated the same process (touching the charged metal sphere A to the neutral plastic sphere and then separating them), what charge would you expect on the plastic sphere after separation? Give a numerical answer if possible and explain your reasoning about charge movement in the insulator. (2 marks)

Working:

Q10. (1 mark) Two point charges, $+3.0 \mu\text{C}$ and $-2.0 \mu\text{C}$, are 0.50 m apart in vacuum. Using Coulomb's law ($k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$), what is the magnitude of the electrostatic force between them?

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- A. 0.22 N
- B. 0.022 N
- C. 2.2 N
- D. 0.0022 N

Q11. (3 marks) Explain what is meant by the statement "the electric field is a characteristic of a charge distribution". In your answer, state why the electric field at a point does not depend on the test charge used to measure it, and give one brief physical example (with formula or units) to illustrate.

Q12. (4 marks) Two point charges lie on the x-axis: a charge $+3.0 \mu\text{C}$ is at $x = 0.00 \text{ m}$ and a charge $-1.0 \mu\text{C}$ is at $x = +0.20 \text{ m}$. Calculate the net electric field (magnitude and direction) at the point $x = +0.10 \text{ m}$. Take $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$. (4 marks)

Working:

Q13. (1 mark) Which description correctly summarizes the standard rules for drawing electric field lines for static charge distributions?

- A. Field lines are drawn so their tangent at any point shows the field direction; the number of lines per area indicates field strength; lines start on positive charges and end on negative charges.
- B. Field lines always cross where the field is strongest; arrows on lines show the direction a positive test charge will move and the spacing of lines is arbitrary.
- C. Field lines are circular around every charge; their density indicates the speed a charged particle will have and they always begin on negative charges.
- D. Field lines are drawn perpendicular to equipotential surfaces only at infinity; they neither indicate direction nor relative strength but only show connectivity between charges.

Q14. (3 marks) A uniform electric field of magnitude $E = 5.0 \times 10^3 \text{ N/C}$ is directed at an angle of 30.0° to the normal of a small flat square surface element of area $\Delta S = 2.0 \text{ cm}^2$. (a) Calculate the electric flux through this surface element. (b) Briefly state the physical meaning of the sign and magnitude of the flux in terms of field lines. (Total 3 marks)

Q15. (4 marks) A uniform electric field of magnitude 200 N C^{-1} points in the $+x$ direction. A small square plate of area 0.020 m^2 is placed so that the unit normal to the plate makes an angle of 30.0° with the $+x$ direction. Calculate the electric flux through the plate. Give the answer with the correct sign and appropriate SI units. (4 marks)

Working:

Q16. (1 mark) A permanent electric dipole with dipole moment $p = 4.0 \text{ C}\cdot\text{m}$ is placed in a uniform electric field of magnitude $E = 2.0 \text{ N/C}$ so that p makes an angle $\theta = 30^\circ$ with the field. Which statement correctly describes the torque on the dipole and any net force it experiences?

- A. The dipole experiences a torque of magnitude $\tau = pE \sin\theta = (4.0 \text{ C}\cdot\text{m})(2.0 \text{ N/C}) \sin 30^\circ = 4.0 \text{ N}\cdot\text{m}$ and no net force.

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B. The dipole experiences a torque of magnitude $\tau = pE \cos\theta = (4.0 \text{ C}\cdot\text{m})(2.0 \text{ N/C}) \cos 30^\circ \sim 6.93 \text{ N}\cdot\text{m}$ and no net force.

- C. The dipole feels no torque and a net force of magnitude $F = pE = (4.0 \text{ C}\cdot\text{m})(2.0 \text{ N/C}) = 8.0 \text{ N}$ toward the field direction.
- D. The dipole experiences a torque of magnitude $\tau = pE = (4.0 \text{ C}\cdot\text{m})(2.0 \text{ N/C}) = 8.0 \text{ N}\cdot\text{m}$ and a net force of zero.

Q17. (3 marks) Two point charges are in the xy-plane: $q_1 = +2.0 \mu\text{C}$ at $\mathbf{r}_1 = (0.00 \mathbf{i} + 0.03 \mathbf{j} + 0.00 \mathbf{k}) \text{ m}$ and $q_2 = -3.0 \mu\text{C}$ at $\mathbf{r}_2 = (0.04 \mathbf{i} + 0.00 \mathbf{j} + 0.00 \mathbf{k}) \text{ m}$. Using Coulomb's law in vector form, calculate the electrostatic force on q_1 due to q_2 . Give the force as a vector with units and state briefly whether the force is attractive or repulsive. ($k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$)

Q18. (4 marks) Three point charges lie on the x-axis: $q_1 = +5.0 \mu\text{C}$ at $x = 0.00 \text{ m}$, $q_2 = -3.0 \mu\text{C}$ at $x = 0.20 \text{ m}$, and $q_3 = +2.0 \mu\text{C}$ at $x = -0.15 \text{ m}$. Using Coulomb's law and the principle of superposition, calculate the net electrostatic force on q_2 . Give the answer as a vector along the x-axis (include magnitude, direction, and units). Use $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.

Working:

Q19. (1 mark) Using Gauss's law, which set of electric field expressions correctly gives the magnitude of the field (with SI units) for each charged object: (1) at distance r from an infinitely long straight wire with linear charge density λ , (2) near an infinite uniformly charged plane with surface charge density σ , and (3) at distance r from the centre of a uniformly charged thin spherical shell of total charge q and radius R (for r both $< R$ and $> R$)?

- A. Wire: $E = \lambda/(2\pi \epsilon_0 r)$ directed radially (N/C). Plane: $E = \sigma/(2\epsilon_0)$ normal to the plane (N/C). Sphere: for $r > R$, $E = q/(4\pi\epsilon_0 r^2)$ radially; for $r < R$, $E = 0$ (N/C).
- B. Wire: $E = \lambda/(4\pi \epsilon_0 r)$ radially (N/C). Plane: $E = \sigma/(\epsilon_0)$ normal to the plane (N/C). Sphere: for $r > R$, $E = q/(4\pi\epsilon_0 r^2)$ radially; for $r < R$, $E = 0$ (N/C).
- C. Wire: $E = \lambda/(2\pi \epsilon_0 r)$ directed radially (N/C). Plane: $E = \sigma/(2\epsilon_0)$ normal to the plane (N/C). Sphere: for $r > R$, $E = q/(4\pi\epsilon_0 r^2)$ radially; for $r < R$, $E = q/(4\pi\epsilon_0 R^2)$ (N/C).
- D. Wire: $E = \lambda/(2\pi \epsilon_0 r)$ directed radially (N/C). Plane: $E = \sigma/(\epsilon_0)$ normal to the plane (N/C). Sphere: for $r > R$, $E = q/(4\pi\epsilon_0 r^2)$ radially; for $r < R$, $E = 0$ (N/C).

Q20. (4 marks) Two neutral insulating objects A and B are rubbed together and, as a result, 5.0×10^{12} electrons are transferred from object A to object B. (a) Calculate the electric charge on object A and on object B after the rubbing. (b) Calculate the net charge of the two-object system and explain, using atomic-scale charge transfer and the quantisation of charge, why rubbing transfers electrons but does not create net charge. (Use the electron charge magnitude $e = 1.60 \times 10^{-19} \text{ C}$.)

Working:

Q21. (1 mark) A simple electric dipole consists of charges $+q$ and $-q$ separated by $2a$ along the z-axis, so the dipole moment is $\mathbf{p} = 2aq \hat{\mathbf{z}}$ (pointing from $-q$ to $+q$). For points far from the dipole

($r \gg a$) give the leading-order expressions (magnitude, sign and direction) for the electric field on (1) the dipole axis at distance r measured from the dipole center along $+z$ and (2) a point in the equatorial plane at distance r from the center. Use SI units and write your answers in terms of p and r with the Coulomb constant $1/(4\pi\epsilon_0)$.

- A. On axis: $E = (1/4\pi\epsilon_0) (2p / r^3) \hat{z}$ (pointing away from the center toward $+q$). Equatorial plane: $E = - (1/4\pi\epsilon_0) (p / r^3) \hat{z}$ (pointing opposite to the dipole moment).
- B. On axis: $E = (1/4\pi\epsilon_0) (p / r^2) \hat{z}$. Equatorial plane: $E = - (1/4\pi\epsilon_0) (p / r^2) \hat{z}$.
- C. On axis: $E = (1/4\pi\epsilon_0) (2p / r^3) \hat{r}$ where $\hat{r}$ points radially outward from the center; Equatorial plane: $E = (1/4\pi\epsilon_0) (p / r^3) \hat{r}$ pointing outward in the plane.
- D. On axis: $E = - (1/4\pi\epsilon_0) (2p / r^3) \hat{z}$ (pointing toward the center). Equatorial plane: $E = (1/4\pi\epsilon_0) (p / r^3) \hat{z}$ (pointing along the dipole moment).

Q22. (4 marks) Define the linear charge density λ , the surface charge density σ , and the volume charge density ρ for continuous charge distributions. For each density, give the mathematical definition (as charge per unit length/area/volume using macroscopic charge element notation) and state its SI unit.

Working:

Q23. (4 marks) a) Give the operational definition of the electric field at a point in space (what experiment you would perform and how you would calculate the field). State the SI unit of the electric field.

b) A point charge $Q = +5.0 \mu\text{C}$ is placed at the origin. Using your operational definition, calculate the magnitude and direction of the electric field at a point 0.20 m from the origin.

Working:

Q24. (4 marks) Two point charges, $+3.0 \mu\text{C}$ and $-1.0 \mu\text{C}$, are fixed 20.0 cm apart on a horizontal line. Find the magnitude and direction of the electric field at the point midway between them. State the result with appropriate units and show your working.

Working:

Answer Key — teacher copy, not for distribution to students

Answer Key

Q1.

Answer: $F_{\text{on}_2} = (-0.129 \text{ N}) \mathbf{i} + (-0.173 \text{ N}) \mathbf{j}$. Magnitude $|F_{\text{on}_2}| = 0.216 \text{ N}$, directed from q_2 toward q_1 (along the vector $-0.6 \mathbf{i} - 0.8 \mathbf{j}$).

Working: Step 1 — Convert charges to coulombs: $q_1 = +3.0 \mu\text{C} = +3.0 \times 10^{-6} \text{ C}$, $q_2 = -2.0 \mu\text{C} = -2.0 \times 10^{-6} \text{ C}$.

Step 2 — Find displacement from q_2 to q_1 : $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2 = (0.00 - 0.30, 0.00 - 0.40, 0.00 - 0.00) \text{ m} = (-0.30, -0.40, 0.00) \text{ m}$.

Step 3 — Compute separation distance $r = \sqrt{(-0.30)^2 + (-0.40)^2} \text{ m} = \sqrt{0.09 + 0.16} \text{ m} = \sqrt{0.25} \text{ m} = 0.50 \text{ m}$.

Step 4 — Unit vector from q_2 toward q_1 : $\hat{\mathbf{r}} = \mathbf{r} / r = (-0.30/0.50, -0.40/0.50, 0) = (-0.6, -0.8, 0)$.

Step 5 — Use Coulomb's constant $k = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$ and Coulomb's law in vector form:

$$\mathbf{F}_{\text{on } 2} = k q_2 q_1 / r^2 \cdot \hat{\mathbf{r}}.$$

$$\text{Compute } q_1 q_2 = (3.0 \times 10^{-6} \text{ C})(-2.0 \times 10^{-6} \text{ C}) = -6.0 \times 10^{-12} \text{ C}^2.$$

$$\text{Compute denominator } r^2 = (0.50 \text{ m})^2 = 0.25 \text{ m}^2.$$

$$\text{Magnitude prefactor} = k |q_1 q_2| / r^2 = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(6.0 \times 10^{-12} \text{ C}^2) / 0.25 \text{ m}^2 = (8.99 \times 10^9)(2.4 \times 10^{-11}) \text{ N} = 0.21576 \text{ N}.$$

The sign from $q_1 q_2$ is negative, indicating the force on q_2 is toward q_1 , consistent with $\hat{\mathbf{r}}$ pointing toward q_1 .

Thus vector force:

$$\mathbf{F}_{\text{on } 2} = (-0.21576 \text{ N}) \cdot \hat{\mathbf{r}} = (-0.21576)(-0.6, -0.8, 0) \text{??} \text{ (Note: the negative sign is already in } q_1 q_2; \text{ applying correctly gives } \mathbf{F} = (-0.21576)(-0.6, -0.8, 0) \text{ would be positive — instead handle sign carefully below.)}$$

Better: take magnitude 0.21576 N and direction $\hat{\mathbf{r}}$ (which already points from q_2 to q_1). Because $q_1 q_2$ is negative the force is along $\hat{\mathbf{r}}$ (toward q_1). Therefore components:

$$F_x = 0.21576 \times (-0.6) = -0.129456 \text{ N} \rightarrow -0.129 \text{ N (3 s.f.)}$$

$$F_y = 0.21576 \times (-0.8) = -0.172608 \text{ N} \rightarrow -0.173 \text{ N (3 s.f.)}$$

$$\text{Magnitude } |\mathbf{F}_{\text{on } 2}| = 0.21576 \text{ N} \rightarrow 0.216 \text{ N (3 s.f.)}$$

Final answer: $\mathbf{F}_{\text{on } 2} = (-0.129 \text{ N}) \mathbf{i} + (-0.173 \text{ N}) \mathbf{j}$, magnitude 0.216 N directed from q_2 toward q_1 .

Rubric — Correct vector electrostatic force on q_2 due to q_1 , with units and clear use of Coulomb's law.

Full credit (4 marks): All steps shown: charges converted to coulombs, displacement and distance computed correctly ($r = 0.50 \text{ m}$), correct unit vector $(-0.6, -0.8, 0)$, correct magnitude 0.216 N to three significant figures, and correct components $F_x = -0.129 \text{ N}$, $F_y = -0.173 \text{ N}$ with units.

Partial credit (2 marks): Most steps correct but with one of the following errors: correct magnitude but one component sign or direction incorrect; or correct components but magnitude not calculated/rounded properly; or units missing. Work must show Coulomb's law and use of $\hat{\mathbf{r}}$. Award 2 marks.

No credit (0 marks): Incorrect or no application of Coulomb's law, wrong distance or unit vector, or numerical result unrelated to shown working. No clear steps connecting Coulomb's law to a vector answer.

Q2.

Correct: A. 0.75 N directed to the left (negative x-direction)

A. Correct. Compute each force with Coulomb's law: F_1 from q_1 : $k(1.0 \times 10^{-6} \text{ C})(2.0 \times 10^{-6} \text{ C}) / (0.20 \text{ m})^2 = 0.4495 \text{ N}$ directed left (repulsion). F_2 from q_2 : $k(1.0 \times 10^{-6} \text{ C})(3.0 \times 10^{-6} \text{ C}) / (0.30 \text{ m})^2 = 0.2997 \text{ N}$ directed left (attraction). Vector sum $= 0.4495 + 0.2997 = 0.7492 \text{ N} \sim 0.75 \text{ N}$ left.

B. Incorrect direction. This option has the correct magnitude but the wrong sign. Both individual forces point to the left (q_0 is repelled by $+q_1$ to the left and attracted to $-q_2$ to the left), so the net force cannot be to the right.

C. Magnitude error from using only the contribution of q_2 or miscomputing one term. 0.30 N is approximately the magnitude of the force from q_2 alone; it ignores or omits the additional $\sim 0.45 \text{ N}$ from q_1 , violating superposition.

D. Magnitude error from using only the contribution of q_1 or arithmetic mistake. 0.45 N is approximately the force from q_1 alone (0.4495 N) but omits the $\sim 0.30 \text{ N}$ from q_2 , so it fails to sum both vector contributions.

Q3.

Answer: (a) Infinite straight wire (linear charge density λ): Choose a cylindrical Gaussian surface of radius r and length L coaxial with the wire. By symmetry E is radial and constant on the curved surface; flux = $E(2\pi rL)$. Enclosed charge = λL . Gauss's law: $E(2\pi rL) = \lambda L/\epsilon_0 \Rightarrow E = \lambda/(2\pi \epsilon_0 r)$, directed radially outward. Units: $C\ m^{-1} / (\epsilon_0 \cdot m) = N\ C^{-1}$.

(b) Infinite plane (surface charge density σ): Choose a pillbox (cylinder) of area A penetrating the plane so it has two flat faces, one on each side. Field is normal and equal in magnitude on both sides, and negligible on the curved side. Flux = $E \cdot A + E \cdot A = 2EA$. Enclosed charge = σA . Gauss's law: $2EA = \sigma A/\epsilon_0 \Rightarrow E = \sigma/(2\epsilon_0)$, directed normal to the plane (same magnitude on both sides). Units: $C\ m^{-2}/(\epsilon_0) = N\ C^{-1}$.

(c) Thin spherical shell (total charge q , radius R): Use a concentric spherical Gaussian surface of radius r .

- For $r > R$ (outside): flux = $E(4\pi r^2)$. Enclosed charge = q . Gauss's law: $E(4\pi r^2) = q/\epsilon_0 \Rightarrow E = q/(4\pi\epsilon_0 r^2)$, as if all charge were at the centre, radial outward. Units: $C/(\epsilon_0 \cdot m^2) = N\ C^{-1}$.

- For $r < R$ (inside): Gaussian surface encloses no charge (enclosed charge = 0 because shell is thin). Thus flux = $E(4\pi r^2) = 0 \Rightarrow E = 0$ inside the shell.

Working: (a) Wire:

1. Choose cylindrical Gaussian surface radius r , length L centered on wire.
2. By cylindrical symmetry E is radial and constant on curved surface; contributions on end caps are zero.
3. Electric flux $\Phi = \text{closed integral } E \cdot dA = E(2\pi rL)$.
4. Enclosed charge $q_{\text{enc}} = \lambda L$.
5. Gauss's law: $\Phi = q_{\text{enc}}/\epsilon_0 \Rightarrow E(2\pi rL) = \lambda L/\epsilon_0$.
6. Solve: $E = \lambda/(2\pi \epsilon_0 r)$. Units: λ ($C\ m^{-1}$) divided by ϵ_0 ($C^2\ N^{-1}\ m^{-2}$) and length r (m) yields $N\ C^{-1}$.

(b) Infinite plane:

1. Choose pillbox crossing plane with flat faces area A ; symmetry gives E normal and same magnitude on both sides.
2. Flux $\Phi = E \cdot A$ (top) + $E \cdot A$ (bottom) = $2EA$.
3. Enclosed charge $q_{\text{enc}} = \sigma A$.
4. Gauss's law: $2EA = \sigma A/\epsilon_0 \Rightarrow E = \sigma/(2\epsilon_0)$.
5. Direction: normal to plane, same magnitude on both sides. Units: σ ($C\ m^{-2}$) over ϵ_0 gives field in $N\ C^{-1}$.

(c) Thin spherical shell:

1. Choose spherical Gaussian surface radius r concentric with shell.
2. By symmetry E is radial and constant on sphere: flux $\Phi = E(4\pi r^2)$.
3. If $r > R$: $q_{\text{enc}} = q$, so $E(4\pi r^2) = q/\epsilon_0 \Rightarrow E = q/(4\pi\epsilon_0 r^2)$.
4. If $r < R$: $q_{\text{enc}} = 0$ (no charge enclosed inside shell), so $E(4\pi r^2) = 0 \Rightarrow E = 0$.
5. Units: q (C) divided by ϵ_0 ($C^2\ N^{-1}\ m^{-2}$) and r^2 (m^2) yields $N\ C^{-1}$.

Rubric — Correct use of Gauss's law and appropriate Gaussian surface to derive each requested electric field expression, with units and direction stated.

Full credit (3 marks): All three cases correctly derived: (a) cylindrical surface leading to $E = \lambda/(2\pi \epsilon_0 r)$ with radial direction and units; (b) pillbox giving $E = \sigma/(2\epsilon_0)$ with direction and units; (c) spherical surface giving $E = q/(4\pi\epsilon_0 r^2)$ for $r > R$ and $E = 0$ for $r < R$, with units. Each derivation shows chosen Gaussian surface, flux = $q_{\text{enc}}/\epsilon_0$ step, and algebraic solution for E .

Partial credit (1 marks): One or two cases correctly derived with appropriate Gaussian surface, flux equation,

and solved expression (with units). Missing or incorrect derivation for the remaining case(s), or correct final formulas given without one clear flux/enclosed-charge step.

No credit (0 marks): No correct derivations; final expressions absent or incorrect for all three cases, or Gauss's law/flux steps not used at all.

Q4.

Answer: Rubbing brings surfaces into close contact so that electrons can move from atoms on one surface to atoms on the other. Which way electrons move depends on the relative tendency of the materials' surface atoms to hold electrons (electron affinity or work function). For example, plastic tends to gain electrons from hair, so individual electrons (each with charge $-e$ where $e = 1.602 \times 10^{-19} \text{ C}$) move from hair atoms to the comb. Each electron removed from the hair leaves behind a positive ion (a deficit of one electron) and each electron gained by the comb produces an extra negative charge. Because charge is quantised, every transferred amount is an integer multiple of e : $q_{\text{transferred}} = n \cdot e$ for some integer n (n is the number of electrons moved). Conservation of charge requires that the total charge of the isolated two-object system does not change during contact and rubbing: every electron that leaves the hair appears on the comb, so the negative charge gained by the comb equals in magnitude the positive charge left on the hair. Thus rubbing redistributes charge between the two bodies but does not create net charge; the sum of their charges before and after rubbing is the same. As an illustrative number: if 10^{12} electrons transfer, the comb gains $-n e = -(10^{12})(1.602 \times 10^{-19} \text{ C}) = -1.60 \times 10^{-7} \text{ C}$ and the hair gains $+1.60 \times 10^{-7} \text{ C}$, leaving the total unchanged.

Rubric — Explanation combining atomic-scale electron transfer, conservation of charge, and quantisation (with an example or numerical illustration).

Full credit (6 marks): Explains that rubbing allows electrons to move between surface atoms (states reason: electron affinity/work function difference), states that each transferred particle is an electron of charge $-e$ and that transferred charge is $q = n \cdot e$, and explicitly applies conservation of charge to show the negative charge gained by one object equals the positive charge left on the other; includes a correct numerical illustration using $e = 1.602 \times 10^{-19} \text{ C}$ or an equivalent integer-multiple statement. All key points correct and clearly connected.

Partial credit (3 marks): Identifies that electrons move from one object to the other and mentions either conservation of charge or quantisation but not both fully; gives a correct qualitative account that charge is redistributed (not created) but lacks a clear integer/electron explanation or a numerical/quantisation statement, or contains a minor omission or small inaccuracy in connecting the principles.

Minimal / no credit (0 marks): Gives an incorrect or very incomplete explanation (e.g. states rubbing 'creates charge' or only describes macroscopic observation without atomic mechanism), does not mention conservation or quantisation, or contains fundamental errors about where charge comes from.

Q5.

Answer: (a) On the positive z-axis ($\theta = 0$): $E = (1/4\pi\epsilon_0) (2 p / r^3) \hat{z}$.

(b) On the equatorial plane ($\theta = 90^\circ$): $E = -(1/4\pi\epsilon_0) (p / r^3) \hat{z}$.

Working: (a) On-axis (point at $z = r$, $r \gg a$):

- Electric field is superposition of fields from $+q$ at $z = +a$ and $-q$ at $z = -a$, both along $\hat{z}$.
- Field from $+q$ at distance $r-a$: $E_+ = (1/4\pi\epsilon_0) q / (r-a)^2$ in $+\hat{z}$ direction.
- Field from $-q$ at distance $r+a$: $E_- = (1/4\pi\epsilon_0) (-q) / (r+a)^2$ in $+\hat{z}$ direction (because $-q$ produces field toward it).
- Net $E_z = (1/4\pi\epsilon_0) [q/(r-a)^2 - q/(r+a)^2]$
- Expand for $r \gg a$ using binomial approximation: $1/(r \pm a)^2 \sim 1/r^2 (1 \pm 2a/r + \dots)$.

So $q[(1/(r-a)^2) - (1/(r+a)^2)] \sim q[(1/r^2)(1+2a/r) - (1/r^2)(1-2a/r)] = q(4a/r^3)$.

- Thus $E_z \sim (1/4\pi\epsilon_0)(4a q / r^3) = (1/4\pi\epsilon_0)(2p / r^3)$ since $p = 2a q$. Direction is $+\hat{z}$.

(b) Equatorial plane (point at $x = r, z = 0, r \gg a$):

- Consider vertical (z) components; horizontal components from $+q$ and $-q$ cancel by symmetry.

- Field from $+q$ located at $z = +a$: its z -component is $E_{+z} = (1/4\pi\epsilon_0) q [-a] / (r^2 + a^2)^{3/2}$ (negative because it points toward $+q$, i.e. downward).

- Field from $-q$ at $z = -a$: its z -component is $E_{-z} = (1/4\pi\epsilon_0) (-q) [+a] / (r^2 + a^2)^{3/2}$ (also negative contribution).

- Net $E_z = (1/4\pi\epsilon_0) [q(-a) - q(a)] / (r^2 + a^2)^{3/2} = (1/4\pi\epsilon_0) [-2qa] / (r^2 + a^2)^{3/2}$.

- For $r \gg a$, $(r^2 + a^2)^{3/2} \sim r^3$, so $E_z \sim -(1/4\pi\epsilon_0)(2qa / r^3) = -(1/4\pi\epsilon_0)(p / r^3)$.

- Direction is $-\hat{z}$ (toward the negative z -direction).

Rubric — Correct derivation and expression of the dipole far-field on axis and on the equatorial plane in terms of p , using $r \gg a$ approximations.

Full marks (4 marks): Both (a) and (b) derived correctly with clear superposition, appropriate approximations for $r \gg a$, correct algebraic simplification to $E = (1/4\pi\epsilon_0)(2p/r^3) \hat{z}$ on-axis and $E = -(1/4\pi\epsilon_0)(p/r^3) \hat{z}$ on the equator, and units/factors $1/(4\pi\epsilon_0)$ shown.

Partial credit (2 marks): One of the two positions (either axis or equator) derived correctly with correct use of p and approximations, or both derived but with a minor algebraic slip that does not change dependence on p and r (e.g. a factor-of-2 error on one step but final expressions show correct $1/r^3$ scaling and correct sign for at least one case).

No credit (0 marks): Incorrect or missing derivations; missing use of superposition or $r \gg a$ approximation; final expressions do not show dependence on p or give incorrect r -dependence (not proportional to $1/r^3$).

Q6.

Correct: C . λ is charge per unit length (C/m), σ is charge per unit area (C/m²), ρ is charge per unit volume (C/m³).

A. Wrong: swaps the meanings of σ and ρ . Surface density σ should be charge per unit area (C/m²), and volume density ρ should be charge per unit volume (C/m³).

B. Wrong: gives λ the unit of volume density and ρ the unit of length — units for λ and ρ are confused. Linear density λ must be C/m and ρ must be C/m³.

C. Correct: λ = charge per unit length (measured in coulombs per metre, C/m), σ = charge per unit area (C/m²), and ρ = charge per unit volume (C/m³); each definition matches its geometric measure (length, area, volume).

D. Wrong: defines λ incorrectly as charge per area (mixing it with σ). Linear density λ is charge per unit length (C/m), not charge per cross-sectional area.

Q7.

Answer: Operational definition: Place a small positive test charge q_0 at the point and measure the electrostatic force F experienced by that test charge. The electric field vector at that point is defined as $E = F / q_0$ (so E points in the direction of the force on a positive test charge). The SI unit of E is newtons per coulomb (N C⁻¹). As an example, the field of a point charge Q at the origin is $E(r) = (1 / 4\pi\epsilon_0) \cdot Q \hat{r} / r^2$, in N C⁻¹.

Rubric — Correct operational definition (procedure and equation), statement of direction, and correct SI unit

Full marks (3 marks): Clearly states the measurement procedure (place small positive test charge q_0 and measure force F), gives the correct relation $E = F/q_0$ with E as a vector and notes that E points in the direction of the force on a positive test charge, and states the SI unit N C⁻¹ (or N/C). May include the point-charge example

formula.

Partial credit (2 marks): Gives the correct formula $E = F/q_0$ and the SI unit $N\ C^{-1}$ but omits either the explicit measurement procedure (placing a test charge) or the statement about the direction of E for a positive test charge, or presents these parts ambiguously.

Minimal credit (1 marks): Mentions that the electric field exerts a force on a charge or gives the SI unit $N\ C^{-1}$ but does not provide the correct defining relation $E = F/q_0$ (or gives an incorrect relation) and lacks a clear measurement procedure.

Q8.

Answer: Inside a conductor in electrostatic equilibrium the electric field is zero because free charges within the conductor move until there is no net force on them. If any electric field existed in the conducting material, free electrons (or positive charges) would feel a force and move, contradicting equilibrium. To show this using Gauss's law, choose a Gaussian surface entirely inside the conducting material (but within the metal, not in the cavity). Because charges in a conductor in electrostatic equilibrium reside only on surfaces, the net charge enclosed by this Gaussian surface is zero. Gauss's law, closed integral $E \cdot dA = Q_{\text{enclosed}}/\epsilon_0$, then gives closed integral $E \cdot dA = 0$, and by symmetry and continuity of E in the metal this implies $E = 0$ everywhere inside the conductor.

Because the field inside the conducting material is zero, any excess free charge placed on the conductor must move to locations where it can produce no internal field in the metal: those are the conductor surfaces. Therefore excess charge resides on the outer surface of the hollow sphere.

If a point charge is placed inside the cavity (not touching the conductor), the free charges in the conductor rearrange: an induced charge equal in magnitude and opposite in sign to the point charge appears on the inner cavity surface so that the field inside the conductor material remains zero. The conductor as a whole may also acquire an additional charge on its outer surface: the total charge on the outer surface equals the original excess charge placed on the conductor plus any charge on the conductor before insertion, minus the induced inner-surface charge. For an isolated neutral conducting shell with a $+q$ point charge in the cavity, an induced $-q$ appears on the inner surface and $+q$ appears on the outer surface. Outside the sphere (for a spherical conductor) the field is as if all net charge were concentrated at the centre — for a general shaped conductor the external field depends only on the net charge on the conductor and not on the position of the charge inside the cavity. The induced inner charge ensures the conductor material has zero internal field, satisfying electrostatic equilibrium and Gauss's law.

Rubric — Correct use of Gauss's law and electrostatic equilibrium to explain zero internal field, surface charge location, and induced charges when a cavity contains a point charge.

Excellent (6 marks) (6 marks): Shows that any internal E would move free charges and so cannot exist; selects a Gaussian surface inside the conductor, states $Q_{\text{enclosed}} = 0$ and applies Gauss's law to conclude $E = 0$ in the metal. Clearly explains that excess charge must therefore be on the outer surface. Describes induced $-q$ on the inner cavity surface when a point charge $+q$ is placed in the cavity (and corresponding $+q$ on the outer surface for an initially neutral shell), and states how the external field depends only on the net charge (for a sphere it is as if charge at the centre).

Partial (3 marks) (3 marks): States that the electric field inside a conductor is zero and that excess charge resides on the surface, but provides only a qualitative argument without correctly using Gauss's law, or gives an

incomplete description of induced charges (e.g. mentions induction but fails to state the equal-and-opposite induced inner charge or its effect on outer charge).

Insufficient (0 marks) (0 marks): Gives incorrect or irrelevant reasoning (e.g. misapplies Gauss's law, claims field inside conductor is nonzero, or fails to connect charge motion to equilibrium), or does not address the effect of a point charge in the cavity.

Q9.

Answer: (a) After contact and separation each identical conducting sphere has $+4.0 \mu\text{C}$.

(b) You should not expect the plastic (insulating) sphere to end up with $+4.0 \mu\text{C}$. Insulators do not allow free movement of charge throughout the whole object, so charge cannot redistribute evenly between the two objects. Only a small, localized amount of charge (if any) may be transferred at the contact point; therefore the plastic sphere will not generally carry $+4.0 \mu\text{C}$ and the exact value depends on microscopic transfer at the contact (not equal sharing).

Working: (a) Conservation of charge: total initial charge = $+8.0 \mu\text{C} + 0 \mu\text{C} = +8.0 \mu\text{C}$. Two identical conducting spheres in contact share charge equally because charges are free to move and the spheres have identical capacitance. Final charge on each = total charge / 2 = $(8.0 \mu\text{C}) / 2 = 4.0 \mu\text{C}$. Units: μC (microcoulombs).

(b) For an insulator, charges are not free to move throughout the object. When the charged metal sphere touches the plastic sphere, there may be only a small, localized transfer of charge at the contact region; charge will not redistribute over the whole plastic sphere. Therefore you cannot calculate a simple equal-sharing value like $4.0 \mu\text{C}$; the plastic sphere will not generally end up with $+4.0 \mu\text{C}$. The exact transferred amount depends on surface properties and is not determined by simple equal sharing.

Rubric — Correct calculation of equal charge sharing for conductors and correct explanation of why insulators do not equally share charge

Full marks (4 marks): Part (a): Shows conservation of charge and divides $8.0 \mu\text{C}$ equally to obtain $4.0 \mu\text{C}$ on each conducting sphere. Part (b): Clearly states that an insulator will not allow free charge redistribution so it will not end up with $+4.0 \mu\text{C}$; explains that only localized transfer may occur and equal sharing does not apply.

Partial credit (2 marks): Part (a): Correct numerical result ($4.0 \mu\text{C}$) shown but working is minimal or missing units; Part (b): Offers a vague or partially correct explanation (e.g. says insulator 'may not' share charge but gives no reason about lack of free charge movement).

No credit (0 marks): Part (a): Incorrect calculation (not $4.0 \mu\text{C}$) or violates conservation of charge. Part (b): Incorrect statement that the insulator will share charge equally or gives no relevant explanation about charge mobility.

Q10.

Correct: A. 0.22 N

A. Correct: $F = k |q_1 q_2| / r^2 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C} \times 2.0 \times 10^{-6} \text{ C}) / (0.50 \text{ m})^2 = (8.99 \times 10^9)(6.0 \times 10^{-12}) / (0.25) \sim 0.216 \text{ N}$, rounded to 0.22 N . Units N are included.

B. Mistake: result is ten times too small. A student choosing this likely divided by an extra factor of 10 (decimal-place error) or incorrectly handled powers of ten when multiplying the charges.

C. Mistake: result is ten times too large. This distractor reflects treating r as 0.50 but forgetting to square it (or misplacing a decimal), giving $F \sim 2.16 \text{ N}$ instead of 0.216 N .

D. Mistake: result is 100 times too small. This choice indicates the student may have squared the charges or introduced an extra factor of 10^{-2} when converting microcoulombs, producing an incorrect power-of-ten handling.

Q11.

Answer: The electric field is a vector field defined at each point in space that is produced solely by the source charges (the charge distribution). It describes the electrical environment created by those source charges independent of any measurement device. The field E at a point determines the force a test charge will feel, via $F = q_{\text{test}} E$, so the field is the property of the sources while the force depends on the value of the test charge. Therefore the value of E is independent of which test charge you place there; changing the test charge changes the force but not E . Example: a single point charge Q at the origin produces an electric field $E(r) = (1/(4\pi\epsilon_0)) \cdot (Q / r^2) \hat{r}$ (units N C^{-1} or V m^{-1}). If you place a test charge q_1 or q_2 at distance r , the forces are $F_1 = q_1 E$ and $F_2 = q_2 E$, showing E is the same for both test charges while the forces scale with q .

Rubric — Explanation that the electric field is produced by source charges and is independent of the test charge, supported by $F = qE$ and an illustrative example with units or formula.

3 marks (3 marks): Complete explanation: defines the electric field as a vector field produced by the source charge distribution, explicitly states that E is independent of the test charge and justifies this using $F = q_{\text{test}} E$, and gives a correct illustrative example (e.g. point charge formula) including appropriate units or notation.

2 marks (2 marks): Mostly correct: gives a clear definition that E is produced by source charges and states independence from the test charge, and either cites $F = qE$ or gives a correct example (but not both fully developed).

1 mark (1 marks): Partial understanding: makes a relevant statement such as ' E characterises the source charges' or 'the force depends on the test charge' but does not clearly connect these ideas or give a correct example or formula.

Q12.

Answer: Net electric field = $3.60 \times 10^6 \text{ N/C}$ directed to the +x (right).

Working: Step 1: Position and distances. The field point is at $x = 0.10 \text{ m}$. Distance from $+3.0 \mu\text{C}$ at $x=0.00 \text{ m}$ is $r_1 = 0.10 \text{ m}$. Distance from $-1.0 \mu\text{C}$ at $x=0.20 \text{ m}$ is $r_2 = 0.10 \text{ m}$.

Step 2: Electric field magnitude from each charge using $E = k|q|/r^2$.

For $q_1 = +3.0 \mu\text{C} = +3.0 \times 10^{-6} \text{ C}$:

$$\begin{aligned} E_1 &= k|q_1|/r_1^2 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C})/(0.10 \text{ m})^2 \\ &= (8.99 \times 10^9)(3.0 \times 10^{-6})/(0.01) \text{ N/C} \\ &= (8.99 \times 10^9)(3.0 \times 10^{-4}) \text{ N/C} \\ &= 2.697 \times 10^6 \text{ N/C}. \end{aligned}$$

Direction: q_1 is positive, so field at a point to its right is away from $q_1 \rightarrow$ to the +x direction.

For $q_2 = -1.0 \mu\text{C} = -1.0 \times 10^{-6} \text{ C}$:

$$\begin{aligned} E_2 &= k|q_2|/r_2^2 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(1.0 \times 10^{-6})/(0.10^2) \\ &= (8.99 \times 10^9)(1.0 \times 10^{-4}) \text{ N/C} \\ &= 8.99 \times 10^5 \text{ N/C}. \end{aligned}$$

Direction: q_2 is negative, so the field points toward q_2 . The field point is to the left of q_2 , so the field at the point is to the +x direction.

Step 3: Superpose (vector sum). Both fields point in +x direction, so magnitudes add:

$$E_{\text{net}} = E_1 + E_2 = 2.697 \times 10^6 \text{ N/C} + 8.99 \times 10^5 \text{ N/C} = 3.596 \times 10^6 \text{ N/C}.$$

Step 4: Report with suitable sig figs.

$E_{\text{net}} \sim 3.60 \times 10^6 \text{ N/C}$ to the +x (right).

Rubric — Application of the superposition principle and correct vector addition to compute the net electric field (magnitude and direction, with units).

Full marks (4 marks): Correctly computes both individual field magnitudes with units, determines both field directions correctly, adds them using superposition to give $E_{\text{net}} \sim 3.60 \times 10^6 \text{ N/C}$ to the +x (right); all arithmetic shown and units carried through.

Partial credit (2 marks): Correct setup and computation of the two individual field magnitudes (with units) and their directions, but contains an arithmetic or rounding error in the final sum OR gives the correct numeric result but omits units or direction; shows most working.

Minimal / no credit (0 marks): Incorrect use of superposition (e.g. subtracts magnitudes without considering directions), or wrong distances used, or missing essential steps; final answer incorrect or not given.

Q13.

Correct: A. Field lines are drawn so their tangent at any point shows the field direction; the number of lines per area indicates field strength; lines start on positive charges and end on negative charges.

A. Correct — this states the three standard rules: (1) tangent gives local direction of E, (2) density of lines per unit area perpendicular to lines represents field magnitude, and (3) lines originate on positive and terminate on negative charges.

B. This is incorrect because field lines do not cross (crossing would imply two different directions of E at one point); arrows do show the direction a positive test charge would feel a force but the spacing is not arbitrary — it represents field strength.

C. This is incorrect because field lines are not necessarily circular around every charge (they are radial for isolated point charges); density represents field strength, not particle speed, and lines start on positive charges, not negative.

D. This is incorrect because field lines are perpendicular to equipotentials at every point (not only at infinity) and they do indicate both direction and relative strength, not merely connectivity.

Q14.

Answer: (a) Convert area to SI units: $\Delta S = 2.0 \text{ cm}^2 = 2.0 \times 10^{-4} \text{ m}^2$. Use $\Phi = E \cdot \Delta S = E \Delta S \cos \theta$.
 $\Phi = (5.0 \times 10^3 \text{ N/C})(2.0 \times 10^{-4} \text{ m}^2) \cos 30.0^\circ = 1.00 \times \cos 30.0^\circ \text{ N}\cdot\text{m}^2/\text{C} \sim 0.866 \text{ N}\cdot\text{m}^2/\text{C}$.

(b) The positive value means field lines pass through the surface in the direction of the chosen normal. The magnitude $0.866 \text{ N}\cdot\text{m}^2/\text{C}$ measures how many field lines (per unit charge) cross the small area — larger flux means more field lines crossing the element.

Working: Step 1: Convert area to metres squared.

$$\Delta S = 2.0 \text{ cm}^2 \times (1 \times 10^{-2} \text{ m} / 1 \text{ cm})^2 = 2.0 \times 10^{-4} \text{ m}^2.$$

Step 2: Write formula for flux through a surface element with angle θ between field and area normal:

$$\Phi = E \Delta S \cos \theta.$$

Step 3: Substitute numbers (carry units):

$$E = 5.0 \times 10^3 \text{ N/C}, \Delta S = 2.0 \times 10^{-4} \text{ m}^2, \theta = 30.0^\circ.$$

$$\Phi = (5.0 \times 10^3 \text{ N/C}) \times (2.0 \times 10^{-4} \text{ m}^2) \times \cos(30.0^\circ).$$

$$\text{Compute product before cosine: } (5.0 \times 10^3)(2.0 \times 10^{-4}) = 1.00 \text{ (units N}\cdot\text{m}^2/\text{C)}.$$

$$\cos 30.0^\circ = \sqrt{3}/2 \sim 0.866025.$$

$$\Phi = 1.00 \times 0.866025 \text{ N}\cdot\text{m}^2/\text{C} \sim 0.866 \text{ N}\cdot\text{m}^2/\text{C}.$$

Step 4: Interpretation: Positive flux indicates net field lines pass through the surface in the direction of the

chosen normal; magnitude $0.866 \text{ N}\cdot\text{m}^2/\text{C}$ quantifies how many field lines cross the area element (per unit charge).

Rubric — Correct calculation of electric flux and clear physical interpretation of its sign and magnitude.

Full credit (3 marks): Correctly converts area to m^2 , applies $\Phi = E \Delta S \cos\theta$, obtains $\sim 0.866 \text{ N}\cdot\text{m}^2/\text{C}$ with units, and states that the positive value means field lines pass through the surface in the direction of the chosen normal and that magnitude measures how many field lines cross the area.

Partial credit (2 marks): Shows the correct formula and method and gives a numerical answer of the right order (e.g. ~ 0.9 or 0.87) but has a minor arithmetic or unit slip, or gives the correct numerical flux but only a minimal or slightly unclear physical interpretation.

No credit (0 marks): Uses an incorrect formula (e.g. missing $\cos\theta$), gets a number far from the correct value, or gives no meaningful physical interpretation of sign/magnitude.

Q15.

Answer: Electric flux $= \Phi = E A \cos\theta = (200 \text{ N C}^{-1})(0.020 \text{ m}^2) \cos 30.0^\circ = 4.00 \times 0.8660 \text{ N}\cdot\text{m}^2 \text{ C}^{-1} = 3.46 \text{ N}\cdot\text{m}^2 \text{ C}^{-1}$. The flux is positive ($\Phi = +3.46 \text{ N}\cdot\text{m}^2 \text{ C}^{-1}$).

Working: Definition: electric flux through a surface element $= \mathbf{E} \cdot \mathbf{A} = E A \cos\theta$, where θ is angle between $\mathbf{E}$ and the area normal.

Given: $E = 200 \text{ N C}^{-1}$, $A = 0.020 \text{ m}^2$, $\theta = 30.0^\circ$.

Compute product with units: $E A = (200 \text{ N C}^{-1})(0.020 \text{ m}^2) = 4.00 \text{ N}\cdot\text{m}^2 \text{ C}^{-1}$.

Include angle: $\Phi = (4.00 \text{ N}\cdot\text{m}^2 \text{ C}^{-1}) \times \cos 30.0^\circ = 4.00 \times 0.866025\dots \text{ N}\cdot\text{m}^2 \text{ C}^{-1} = 3.4641\dots \text{ N}\cdot\text{m}^2 \text{ C}^{-1}$.

Round to three significant figures: $\Phi = 3.46 \text{ N}\cdot\text{m}^2 \text{ C}^{-1}$.

Sign: $\cos 30^\circ$ is positive, field and normal have acute angle, so flux is positive. Final answer: $\Phi = +3.46 \text{ N}\cdot\text{m}^2 \text{ C}^{-1}$.

Rubric — Correct use of flux formula, substitution with units, correct numerical result and sign.

Full credit (4 marks): Uses $\Phi = E A \cos\theta$, substitutes $E = 200 \text{ N C}^{-1}$, $A = 0.020 \text{ m}^2$, $\theta = 30^\circ$, carries units through, evaluates $\cos 30^\circ$ correctly and gives $\Phi = +3.46 \text{ N}\cdot\text{m}^2 \text{ C}^{-1}$ (or equivalent to 3.46 with correct units and positive sign).

Partial credit (2 marks): Shows correct formula $\Phi = E A \cos\theta$ and substitutes values with units, but arithmetic is incorrect or final value is mis-rounded (gives a value close to 3.46, e.g. 3.5 or 3.4, or omits sign); or uses $E \cdot A$ but forgets the $\cos\theta$ factor (giving $4.00 \text{ N}\cdot\text{m}^2 \text{ C}^{-1}$) — clear working shown earns 2 marks.

No credit (0 marks): No correct formula or substitution shown (e.g., random number or unrelated method), or units entirely missing with no coherent calculation.

Q16.

Correct: A. The dipole experiences a torque of magnitude $\tau = pE \sin\theta = (4.0 \text{ C}\cdot\text{m})(2.0 \text{ N/C}) \sin 30^\circ = 4.0 \text{ N}\cdot\text{m}$ and no net force.

A. Correct: torque on a dipole in a uniform field is $\tau = |\mathbf{p} \times \mathbf{E}| = pE \sin\theta$. Plugging $p = 4.0 \text{ C}\cdot\text{m}$, $E = 2.0 \text{ N/C}$, $\sin 30^\circ = 0.5$ gives $\tau = 4.0 \text{ N}\cdot\text{m}$. A uniform field exerts equal and opposite forces on the charges so the net force is zero. Units: $\text{C}\cdot\text{m}\cdot\text{N/C} = \text{N}\cdot\text{m}$.

B. Wrong: this uses $\cos\theta$ instead of $\sin\theta$. The magnitude of $\mathbf{p} \times \mathbf{E}$ depends on the sine of the angle between $\mathbf{p}$ and $\mathbf{E}$, so $\cos 30^\circ$ is not the correct trigonometric factor.

C. Wrong: a uniform field produces no net force on a pure dipole because the forces on the equal and opposite charges cancel; also it ignores the angular dependence and gives the wrong expression for torque (it claims no torque).

D. Wrong: this gives $\tau = pE$ (the value for $\sin\theta = 1$) even though here $\theta = 30^\circ$ so $\tau = pE \sin 30^\circ$; it also correctly states zero net force but the torque magnitude is incorrect.

Q17.

Answer: $F_{\text{on1}} = (17.26 \text{ i} - 12.95 \text{ j} + 0.00 \text{ k}) \text{ N}$. The force is attractive (q_1 is pulled toward q_2).

Working: 1. Write Coulomb's law (vector form) for force on q_1 due to q_2 : $F = k q_1 q_2 (r_1 - r_2) / |r_1 - r_2|^3$.

2. Compute charge product and convert μC to C : $q_1 = 2.0 \mu\text{C} = 2.0 \times 10^{-6} \text{ C}$, $q_2 = -3.0 \mu\text{C} = -3.0 \times 10^{-6} \text{ C}$; $q_1 q_2 = (2.0 \times 10^{-6})(-3.0 \times 10^{-6}) = -6.0 \times 10^{-12} \text{ C}^2$.

3. Compute displacement vector $r_1 - r_2$: $r_1 - r_2 = (0.00 - 0.04, 0.03 - 0.00, 0.00 - 0.00) \text{ m} = (-0.04, 0.03, 0.00) \text{ m}$.

4. Compute separation magnitude: $|r_1 - r_2| = \sqrt{(-0.04)^2 + (0.03)^2} \text{ m} = \sqrt{0.0016 + 0.0009} \text{ m} = \sqrt{0.0025} \text{ m} = 0.05 \text{ m}$.

5. Compute $|r_1 - r_2|^3 = (0.05 \text{ m})^3 = 0.000125 \text{ m}^3$.

6. Compute scalar prefactor: $k q_1 q_2 / |r|^3 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(-6.0 \times 10^{-12} \text{ C}^2) / 0.000125 \text{ m}^3 = (-5.394 \times 10^{-2} \text{ N}\cdot\text{m}^2) / 0.000125 \text{ m}^3 = -431.52 \text{ N/m}$.

(Units reduce to N/m times m gives N when multiplied by r -vector components.)

7. Multiply by r -vector to get force: $F = (-431.52 \text{ N/m}) \times (-0.04, 0.03, 0.00) \text{ m} = (17.2608, -12.9456, 0.0000) \text{ N}$.

8. Round to three significant figures: $F \sim (17.26 \text{ i} - 12.95 \text{ j} + 0.00 \text{ k}) \text{ N}$.

9. Sign/physical interpretation: q_2 has opposite sign to q_1 , so the force on q_1 is attractive; the vector points from q_1 toward q_2 (which matches the computed components).

Rubric — Correct vector force computation (components with units) and correct statement of attraction/repulsion.

Full marks (3 marks): Gives the correct vector force components with units ($\sim(17.26 \text{ i} - 12.95 \text{ j} + 0.00 \text{ k}) \text{ N}$ or equivalent to three significant figures) and states that the force is attractive, with evidence of correct direction.

Partial credit (2 marks): Gives the correct magnitude and direction (e.g. correct components up to a sign or one component omitted) or the correct vector but missing units, and indicates attraction or repulsion (may be implicit).

Minimal credit (1 marks): Shows correct setup: correct $r_1 - r_2$, separation magnitude, or scalar prefactor, but arithmetic error or missing final vector; or gives only a qualitative statement about attraction/repulsion without a correct numeric vector.

Q18.

Answer: Net force on $q_2 = 3.81 \text{ N}$ toward the negative x -direction, i.e. $F_{\text{net}} = -3.81 \text{ i N}$ (to two decimal places).

Working: Step 1: Convert charges to coulombs.

$$q_1 = +5.0 \mu\text{C} = +5.0 \times 10^{-6} \text{ C}$$

$$q_2 = -3.0 \mu\text{C} = -3.0 \times 10^{-6} \text{ C}$$

$$q_3 = +2.0 \mu\text{C} = +2.0 \times 10^{-6} \text{ C}$$

Step 2: Find distances along x between q_2 and the other charges (use magnitudes).

Distance r_{12} between q_1 ($x=0.00 \text{ m}$) and q_2 ($x=0.20 \text{ m}$): $r_{12} = 0.20 \text{ m}$

Distance r_{32} between q_3 ($x=-0.15 \text{ m}$) and q_2 ($x=0.20 \text{ m}$): $r_{32} = 0.20 - (-0.15) = 0.35 \text{ m}$

Step 3: Compute magnitude of force on q_2 due to q_1 using Coulomb's law (scalar magnitude).

$$F_{12} = k |q_1 q_2| / r_{12}^2 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2) \times (|5.0 \times 10^{-6} \text{ C} \times 3.0 \times 10^{-6} \text{ C}|) / (0.20 \text{ m})^2$$

$$= 8.99 \times 10^9 \times (15.0 \times 10^{-12} \text{ C}^2) / 0.04 \text{ m}^2$$

$$= 8.99 \times 10^9 \times 3.75 \times 10^{-10} \text{ N}$$

$$= 3.37125 \text{ N}$$

Direction: q_1 is positive and q_2 is negative $\Rightarrow$ attraction toward q_1 (which is at smaller x), so force on q_2 is toward negative x . Represent as $F_{12} = -3.37125 \text{ i N}$.

Step 4: Compute magnitude of force on q2 due to q3.

$$\begin{aligned}F_{32} &= k |q_3 q_2| / r_{32}^2 = (8.99 \times 10^9) \times (|2.0 \times 10^{-6} \times 3.0 \times 10^{-6}|) / (0.35)^2 \\&= 8.99 \times 10^9 \times (6.0 \times 10^{-12}) / 0.1225 \\&= 8.99 \times 10^9 \times 4.8980 \times 10^{-11} \text{ N} \\&= 0.4405 \text{ N (approximately)}\end{aligned}$$

Direction: q3 is positive and q2 is negative => attraction toward q3 (which is at smaller x), so force on q2 is toward negative x. Represent as $F_{32} = -0.4405 \text{ i N}$.

Step 5: Superpose (vector sum) the two forces (both along x, both negative):

$$F_{\text{net}} = F_{12} + F_{32} = (-3.37125 - 0.4405) \text{ i N} = -3.81175 \text{ i N}.$$

Step 6: Report with appropriate significant figures (given data to three significant figures):

$F_{\text{net}} \sim -3.81 \text{ i N}$, i.e. 3.81 N toward the negative x-direction.

Rubric — Correct net electrostatic force on q2 using Coulomb's law and vector superposition (magnitude, direction, units, and working).

Full credit (4 marks): Correctly computes both individual forces with k and distances, assigns correct directions (both toward negative x), correctly sums them to give $F_{\text{net}} = -3.81 \text{ i N}$ (or equivalent to two decimal places), and includes units and clear steps.

Partial credit (2 marks): Shows correct method: computes individual force magnitudes and directions but has an arithmetic slip or reports final magnitude with an incorrect sign or wrong rounding that changes the answer by more than the allowed rounding (final answer around 2–4 N toward negative x), or omits clear unit labeling for one step.

No credit (0 marks): Incorrect method (no use of Coulomb's law or superposition), missing vector directions, or final answer is numerically and directionally incorrect with no correct intermediate calculations.

Q19.

Correct: A. Wire: $E = \lambda / (2\pi \epsilon_0 r)$ directed radially (N/C). Plane: $E = \sigma / (2\epsilon_0)$ normal to the plane (N/C). Sphere: for $r > R$, $E = q / (4\pi\epsilon_0 r^2)$ radially; for $r < R$, $E = 0$ (N/C).

A. Correct: derived by Gauss's law. For an infinite straight wire use a cylindrical Gaussian surface to get $E(2\pi rL) = \lambda L / \epsilon_0 \Rightarrow E = \lambda / (2\pi \epsilon_0 r)$. For an infinite plane use a pillbox to get $2EA = \sigma A / \epsilon_0 \Rightarrow E = \sigma / (2\epsilon_0)$. For a thin spherical shell use a spherical surface: for $r > R$ use $E(4\pi r^2) = q / \epsilon_0 \Rightarrow E = q / (4\pi\epsilon_0 r^2)$; for $r < R$ no enclosed charge $\Rightarrow E = 0$. Units are N/C.

B. Incorrect: wire formula is wrong by a factor of 2 — using 4π instead of 2π comes from confusing line symmetry with point symmetry. Plane formula is twice too large; Gauss's law with a pillbox gives $\sigma / (2\epsilon_0)$, not σ / ϵ_0 .

C. Incorrect: wire and plane formulas are correct, but the field inside the spherical shell is not $q / (4\pi\epsilon_0 R^2)$. That value is the field at the surface; inside a thin conducting shell or uniformly charged thin shell the enclosed charge is zero so $E = 0$ for $r < R$.

D. Incorrect: wire and sphere parts are correct, but the plane value is wrong by a factor of 2. A pillbox Gaussian surface shows the plane produces the same magnitude on both sides, giving $\sigma / (2\epsilon_0)$, not σ / ϵ_0 .

Q20.

Answer: (a) Charge transferred = $(5.0 \times 10^{12} \text{ electrons}) \times (1.60 \times 10^{-19} \text{ C per electron}) = 8.0 \times 10^{-7} \text{ C}$. Object A loses these electrons so it gains positive charge $+8.0 \times 10^{-7} \text{ C}$. Object B gains these electrons so it gains negative charge $-8.0 \times 10^{-7} \text{ C}$. (b) The net charge = $(+8.0 \times 10^{-7} \text{ C}) +$

$(-8.0 \times 10^{-7} \text{ C}) = 0 \text{ C}$. At the atomic scale rubbing transfers discrete electrons from atoms on A to atoms on B; no charges are created or destroyed, only redistributed. Because free charge occurs in integer multiples of the electron charge ($q = n e$) the transferred charge is exactly $n e$ with $n = 5.0 \times 10^{12}$, so each object's charge is a whole-number multiple of e . Therefore rubbing moves existing electrons between the objects but does not change the total charge of the isolated system.

Working: Step 1: Number of electrons transferred $n = 5.0 \times 10^{12}$ (given). Step 2: Electron charge $e = 1.60 \times 10^{-19} \text{ C}$ (given). Step 3: Magnitude of transferred charge $q = n \cdot e = (5.0 \times 10^{12}) \times (1.60 \times 10^{-19} \text{ C}) = 8.0 \times 10^{-7} \text{ C}$. Step 4: Object A loses electrons $\rightarrow$ charge on A $= +q = +8.0 \times 10^{-7} \text{ C}$. Step 5: Object B gains electrons $\rightarrow$ charge on B $= -q = -8.0 \times 10^{-7} \text{ C}$. Step 6: Net charge $= (+8.0 \times 10^{-7} \text{ C}) + (-8.0 \times 10^{-7} \text{ C}) = 0 \text{ C}$. Step 7:

Explanation: rubbing removes electrons from atomic orbitals on A and deposits them on B; these are discrete electrons so the transferred charge is an integer multiple of e (quantisation). No charges are created or destroyed in the process, only redistributed, so the net charge remains zero (conservation of charge).

Rubric — Correct numerical calculation of charges and a clear explanation showing conservation of charge and quantisation.

Full credit (4 marks): Correct numerical values for both objects ($A = +8.0 \times 10^{-7} \text{ C}$, $B = -8.0 \times 10^{-7} \text{ C}$) shown with units; correct net charge $= 0 \text{ C}$; explanation explicitly states that electrons are transferred ($n = 5.0 \times 10^{12}$ is an integer multiple of e) and uses conservation of charge to say net charge is unchanged.

Partial credit (2 marks): Calculations produce a magnitude of transferred charge close to $8.0 \times 10^{-7} \text{ C}$ and assigns opposite signs to A and B but either omits units or gives only one object's sign correctly; explanation mentions transfer or conservation but omits quantisation or gives it vaguely.

No credit (0 marks): Numerical answers are incorrect by more than a factor of two, or signs are both the same so conservation is violated; explanation missing or states that rubbing creates net charge (i.e. claims charge is produced rather than transferred).

Q21.

Correct: A. On axis: $E = (1/4\pi\epsilon_0) (2p / r^3) \hat{z}$ (pointing away from the center toward $+q$).

Equatorial plane: $E = - (1/4\pi\epsilon_0) (p / r^3) \hat{z}$ (pointing opposite to the dipole moment).

A. Correct. For $r \gg a$, expand the fields to leading order: on the axis the dipole field is $E = (1/4\pi\epsilon_0)(2p/r^3)$ along $+z$ (away from center toward $+q$), and in the equatorial plane the field is $E = -(1/4\pi\epsilon_0)(p/r^3) \hat{z}$ (toward the dipole axis, opposite the dipole moment). Both have the $1/r^3$ dependence and use $p = 2aq$.

B. Incorrect: This option shows a $1/r^2$ dependence. The far-field dipole field falls off as $1/r^3$, so using $1/r^2$ is the wrong radial scaling (confuses dipole with monopole field).

C. Incorrect: Although the axial expression gives the correct $1/r^3$ scaling, the equatorial field sign/direction is wrong (it should be opposite the dipole moment, i.e., $-\hat{z}$). Also using $\hat{r}$ for axial points is confusing here; the axial result should be written along $\hat{z}$ with the correct sign.

D. Incorrect: Both signs are flipped. On the axis toward $+q$ the field points away from the center (positive $\hat{z}$) so the axial field should be $+2p/(4\pi\epsilon_0 r^3) \hat{z}$, not negative; the equatorial field should be opposite to p ($-p/(4\pi\epsilon_0 r^3) \hat{z}$), not along $+\hat{z}$.

Q22.

Answer: Linear charge density λ : $\lambda = dQ/dl$, where dQ is the charge contained in an infinitesimal length element dl along a charged line. SI unit: coulomb per metre ($\text{C}\cdot\text{m}^{-1}$).

Surface charge density σ : $\sigma = dQ/dA$, where dQ is the charge contained in an infinitesimal area element dA on a charged surface. SI unit: coulomb per square metre ($\text{C}\cdot\text{m}^{-2}$).

Volume charge density ρ : $\rho = dQ/dV$, where dQ is the charge contained in an infinitesimal volume element dV in a charged volume. SI unit: coulomb per cubic metre ($C \cdot m^{-3}$).

Rubric — Correct mathematical definitions and correct SI units for λ , σ , and ρ .

Full credit (4 marks): All three densities given with correct differential definitions ($\lambda = dQ/dl$, $\sigma = dQ/dA$, $\rho = dQ/dV$) and each accompanied by the correct SI unit ($C \cdot m^{-1}$, $C \cdot m^{-2}$, $C \cdot m^{-3}$ respectively).

Partial credit (2 marks): Either (a) all three mathematical definitions correct but one or more SI units missing or incorrect, or (b) two densities correctly defined with correct units while the third is missing or incorrect.

No credit (0 marks): Fewer than two correct definitions and/or units; definitions are incorrect or not given in differential form using dQ and $dl/dA/dV$.

Q23.

Answer: a) Operational definition: Place a small positive test charge q_t (so small that it does not noticeably disturb the source charge distribution) at the point of interest and measure the electric force F that acts on it. The electric field at that point is defined as $E = F / q_t$, directed the same way as the force on a positive test charge. SI unit: newtons per coulomb (N/C).

b) For a point charge Q at the origin, $E(r) = (1 / 4\pi\epsilon_0) \cdot Q / r^2$ directed radially outward for $Q > 0$. Using $k = 1 / (4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$,
 $E = k Q / r^2 = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(5.0 \times 10^{-6} \text{ C}) / (0.20 \text{ m})^2 = 1.12 \times 10^6 \text{ N/C}$, directed radially outward from the origin.

Working: a) Operational procedure and formula:

- Put a test charge q_t (positive and sufficiently small) at the point and measure the force F on it.
- Compute $E = F / q_t$. Units: $[F] = \text{N}$, $[q_t] = \text{C}$ so $[E] = \text{N/C}$.

b) Calculation for the point charge:

- Given: $Q = +5.0 \mu\text{C} = 5.0 \times 10^{-6} \text{ C}$; $r = 0.20 \text{ m}$; $k = 1 / (4\pi\epsilon_0) = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$.
- Use $E(r) = k Q / r^2$.
- Compute $r^2 = (0.20 \text{ m})^2 = 0.0400 \text{ m}^2$.
- Compute numerator: $k Q = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(5.0 \times 10^{-6} \text{ C}) = 8.99 \times 5.0 \times 10^3 \text{ N} \cdot \text{m}^2/\text{C} = 44.95 \times 10^3 \text{ N} \cdot \text{m}^2/\text{C} = 4.495 \times 10^4 \text{ N} \cdot \text{m}^2/\text{C}$.
- Divide: $E = (4.495 \times 10^4 \text{ N} \cdot \text{m}^2/\text{C}) / (0.0400 \text{ m}^2) = 1.12375 \times 10^6 \text{ N/C} \sim 1.12 \times 10^6 \text{ N/C}$.
- Direction: Q is positive, so E is radially outward from the origin.

Rubric — Correct operational definition, correct SI unit, and correct numerical value with direction

Full credit (4 marks): Provides the operational definition mentioning placing a small (non-perturbing) positive test charge, measuring the force F , and computing $E = F/q_t$; states the SI unit N/C; calculates $E = kQ/r^2$ with correct substitution and arithmetic to give $\sim 1.12 \times 10^6 \text{ N/C}$ and states the direction (radially outward).

Partial credit (2 marks): Gives a correct but incomplete operational definition (e.g. says $E = F/q$ without explicitly requiring a small/non-perturbing test charge or omits the measurement step) or states the unit correctly but shows an arithmetic error in the numerical calculation that leads to a result of the correct order of magnitude (marks reduced). Numerical work may show formula and substitution but with one algebraic or arithmetic slip; direction may be omitted or incorrect. Award 2 marks.

No credit (0 marks): Definition incorrect or missing (no mention of measuring force on a test charge or misstates E), unit incorrect, and/or numerical calculation absent or wholly wrong (no reasonable attempt).

Q24.

Answer: Magnitude of electric field at the midpoint: $1.80 \times 10^6 \text{ N/C}$, directed from the positive charge toward the negative charge (i.e. toward the $-1.0 \mu\text{C}$ charge).

Working: Let the charges be $q_1 = +3.0 \mu\text{C} = +3.0 \times 10^{-6} \text{ C}$ (left) and $q_2 = -1.0 \mu\text{C} = -1.0 \times 10^{-6} \text{ C}$ (right).

Separation $d = 20.0 \text{ cm} = 0.200 \text{ m}$, so midpoint is $r = d/2 = 0.100 \text{ m}$ from each charge.

Coulomb constant $k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$.

Magnitude of field from q_1 at midpoint: $E_1 = k |q_1| / r^2 = (8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(3.0 \times 10^{-6} \text{ C}) / (0.100 \text{ m})^2$
 $= (8.99 \times 10^9)(3.0 \times 10^{-6}) / 0.0100$
 $= (8.99 \times 10^9 \times 3.0 \times 10^{-6}) \times 100$
 $= (8.99 \times 3.0 \times 10^3) \text{ N/C}$
 $= 2.697 \times 10^6 \text{ N/C}$.

Direction: q_1 is positive so its field at the midpoint points away from q_1 , i.e. to the right (toward q_2).

Magnitude of field from q_2 at midpoint: $E_2 = k |q_2| / r^2 = (8.99 \times 10^9)(1.0 \times 10^{-6}) / (0.100)^2$
 $= (8.99 \times 10^9 \times 1.0 \times 10^{-6}) / 0.0100$
 $= (8.99 \times 10^9 \times 1.0 \times 10^{-6}) \times 100$
 $= 8.99 \times 10^5 \text{ N/C}$.

Direction: q_2 is negative so its field at the midpoint points toward q_2 , i.e. to the right as well.

Since both fields at the midpoint point to the right, net field magnitude is $E = E_1 + E_2$? (Careful: fields due to positive and negative charges point in same direction here.) Numerically: $E_{\text{net}} = 2.697 \times 10^6 \text{ N/C} + 8.99 \times 10^5 \text{ N/C} = 3.596 \times 10^6 \text{ N/C}$.

However, check signs consistently: if we take right as positive, E_1 (from $+3 \mu\text{C}$) = $+2.697 \times 10^6 \text{ N/C}$, E_2 (from $-1 \mu\text{C}$) = $+8.99 \times 10^5 \text{ N/C}$, so $E_{\text{net}} = +3.596 \times 10^6 \text{ N/C}$ to the right.

But note: the fields above used $r = 0.100 \text{ m}$; if instead one considers directions with respect to each source correctly, the field from the negative charge points toward it (right) with magnitude $8.99 \times 10^5 \text{ N/C}$, and the field from the positive charge points away from it (right) with magnitude $2.697 \times 10^6 \text{ N/C}$. Therefore $E_{\text{net}} = 2.697 \times 10^6 + 8.99 \times 10^5 = 3.596 \times 10^6 \text{ N/C}$ to the right if one had mistakenly taken one as opposite. The correct physical reasoning shows both point to the right, so the correct sum is $3.596 \times 10^6 \text{ N/C}$ to the right. Final answer (corrected with consistent sign and direction): $E_{\text{net}} = 3.60 \times 10^6 \text{ N/C}$, directed toward the $-1.0 \mu\text{C}$ charge (to the right).

Rubric — Correct magnitude (with units) and correct direction of the electric field at the midpoint, using Coulomb's law and distances.

Full credit (4 marks): Correct numerical magnitude ($3.60 \times 10^6 \text{ N/C}$) with units and correct direction (toward the $-1.0 \mu\text{C}$ charge). Working shows $E = k|q|/r^2$ for each charge with $r = 0.100 \text{ m}$, correct arithmetic for both fields, and correct vector addition.

Partial credit (2 marks): Correct method (uses $E = k|q|/r^2$ with $r = 0.100 \text{ m}$ and identifies directions of individual fields) but one arithmetic sign or numerical error that leads to a magnitude differing by about a factor of 2 (e.g. gives $1.80 \times 10^6 \text{ N/C}$) or missing units; direction may be correct or unclear. Working shows most steps.

Minimal/no credit (0 marks): Incorrect method (no use of Coulomb's law at midpoint, wrong distance used), or answer is a number with no units and no supporting working, or direction is opposite and incompatible with shown calculations.